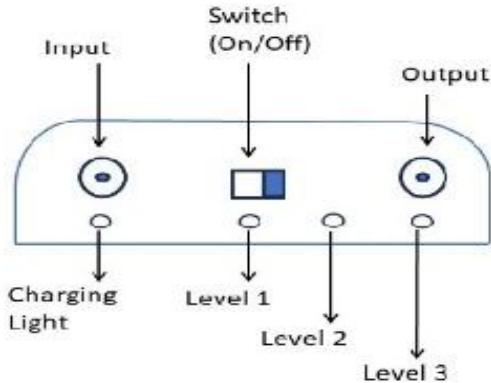
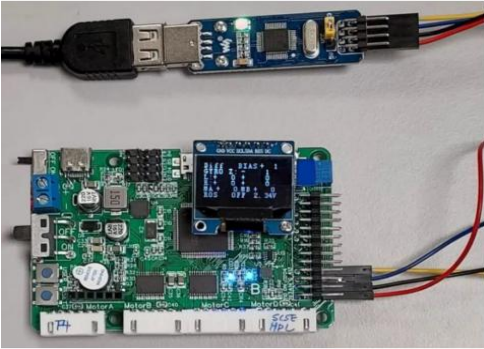
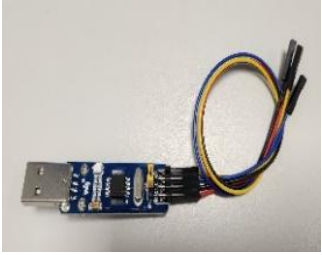
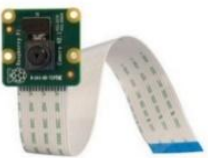
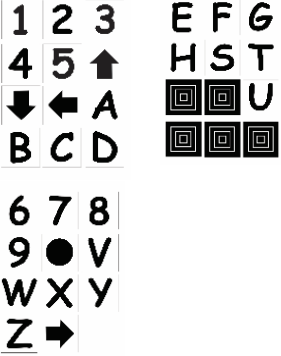


MDP Component List – August 2026

Issue: Team _____ Name: _____ Date: _____	Returned: _____ Name: _____ Date: _____ Remarks: _____ _____
1. Robot car Qty: 1 unit Mounted with the following items:- - Hall Encoder Motors, 12V 1:30 reduction ratio with 2.54mm pitch, 6 pins- 2 units, Model: MG513P3012V - Motor cables – 2pcs mounted - Servo motor 4.8 - 7.4V with rods, Model: HWZ020 - Top acrylic black plate - with 18M3 and 4 M2 holes - Bottom aluminium plate - Battery pack, 12.6V 3400mAH – 1unit	1a. Battery pack, 12.6V 3400mAH - 1 set (for spare) The battery pack contains 3 units of NCR186050B, 3.7V, 3400mAh and Battery box.  1b. Right-angle DC cable with Red/Black leads (connect to the Output port of the battery pack) – 1 pc 
<u>Battery Pack</u>  <u>Power to STM32 board</u> a) Connect Bat Pack Output to STM32 board b) Turn on Battery Pack Switch <u>Charge Battery Pack</u> a) Connect 3pin charger to Battery Pack Input b) Turn On Battery Pack Switch c) Ensure the “Red” Charging Light is on during charging It will disappear after fully charged d) Once Battery Pack is fully charged, the 3-pin charger must turn green.	

2. Resistor – 1Kohm,¼ watt Qty: 1 pc 	3. Resistor –2.2Kohm, ¼ watt Qty: 1 pc 	4. ADC Cable Qty: 1 pc 	
5. Tablet, Samsung Tab 7 Lite Model: SM-T220/Grey Box: _____ S/n : _____ • (Original) USB A to type C cable • (Original) 3-pin USB Power Adaptor • Box Qty: 1 set			6. STM32F407vet6_30D_V2.1 Controller board mounted on acrylic protector with 4 stands Qty: 1 pc 
7. Battery charger Li-ion with 3pin SG safety mark 2A (for Robot) Qty: 1 pc			8. Rpi4b Model B – 4Gb Qty: 1 pc 
9. USB cable type C-C -0.5m Qty: 1 pc			10. Power supply adaptor 5.1V 3A 3pin type C (for Rpi4b) Qty: 1 set 

11. HDMI to micro HDMI cable Qty: 1 pc		12. Sharp GP2Y0A21YK IR Sensor, small, SEN-00020 Qty: 2 pcs	
13. Sensor IR cable Qty: 2 pcs		14. - IR bracket (Orange) Qty: 3 pcs - Bracket (Aluminium) Qty: 1 pc	
Connection for ST-LINK/V2 Serial Wire Debug (SWD) board to STM32F407vet6 Red – 3.3V Black (B) – SWCLK Blue – GND Yellow (A) - SWDIO		15. ▪ ST-LINK/V2, Debugger, USB2.0 – 1 pc ▪ 4-pin header cable (Red/Blk/Blue/Yellow) - 1pc ▪ Extender Cable for debugger (Male USB A to Female USB A)- 1.50m - 1pc	 
16. Ultrasonic Sensor, HCSR04SEN0001 Qty: 1 pc			
17. MicroSD memory card, 32GB Qty: 1 set		18. RPi Camera Module V2 with Flexi cable Qty: 1 pc	

19. • Cable USB A to C, 0.5m Qty: 1 pc • Cable USB A to C, 1.8m Qty: 1 pc		20. Ribbon cable F/F 200mm Qty: 10 pcs	
21. Network cable, 2m Qty: 1 pc		22. Tool Bag - Wire stripper cum cutter - screwdriver (-) - screwdriver (+) - Long nose plier Qty: 1 set	
23. Estimated quantities :- • M2 pillars 10mm M/F, Qty: 20 pcs • M2 screws, 6mm, Qty: 6pcs • M2 nuts, Qty: 6pcs For Rpi mounting		24. Estimated quantities:- • M3 Pillars 25mm F/F Qty: 2 pcs • M3 screws, 6mm, Qty: 4pcs • M3 nuts, Qty: 4pcs For IR camera mounting etc	
25. Obstacle, Small 100X100X150mmh Matt black Qty: 1 pc		26. Card Reader (USB)- Transcend RDF5 Qty: 1 set	
27. Handy storage box with cover • Dimension: L44 X W30.5 X H22 cm Qty: 1 pc		28. Photo images Qty: 1 set of 35 images	

29. Digital Multimeter Model: UT33A+ Box: _____ S/n : _____ • (Original) Red and Black probes • Box Qty: 1 set			
<u>Suppliers for the components:-</u> <u>Singapore Robotic</u> (sensors, motor, <i>IR bracket</i>) Phone: +65-6570 1125, Fax: +65-6462 2051 sales@sgbotic.com		<u>Robot R Us</u> (sensors, motor, <i>IR bracket</i>) sales@robot-r-us.com +65 96569488	
<u>Challenger Technologies Limited</u> (SDcard reader) Tel: 6318 9800, Fax: 63189801		<u>Element14 (Singapore)</u> (Raspberry Pi, Sensors, USB cables, HDMI cables) Phone: +65 6788 0200 , Fax: +65 6788 0300 Email: sg-sales@element14.com	
<u>RS Components Singapore Pte Ltd</u> RPi4b, RPi Camera module V2, usb cables Phone: +65 6865 3400 Fax: +65 6865 3444		<u>BESCOM COMPUTER SYSTEMS PTE LTD</u> (SDcard reader) Tel: 62720808, Fax: 62720909	
<u>Measurement & Metrology (S) Pte Ltd</u> (Digital Multimeter) Tel: 64831691, www.mnm.com.sg		<u>Unicell International Pte Ltd</u> (Battery) Tel: 6742 6330, 6742 6631 Fax: 6742-7763	
<u>Silicon Electronics Company</u> (tools, cables) 10 Jalan Besar #03-02 Sim Lim Tower Tel: 62969006 Fax 2969005		<u>Dream Catcher Technologies Private Limited</u> Tel: +65-62981892 (Robot car parts, STM board)	
<u>TOYOGO SUPERMART (Storage box)</u> 21 Lor 8 Toa Payoh, Toyogo Bld SG 319256 Singapore Phone: +65 62518747			